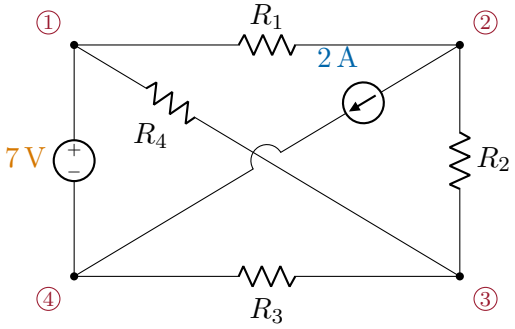
Middle East Technical University
Department of Electrical and Electronics Engineering

EE201 – Circuit Theory I

Example problems and solutions

Node and mesh analysis · Thévenin and Norton equivalents · two-port circuits

Question 1 — a planar circuit: node and mesh analysis



$R_1 = R_2 = R_3 = 1\,\Omega$ and $R_4 = 2\,\Omega$. The resistor R_4 (nodes 1–3) and the 2 A source (from node 2 to node 4) cross without a connection.

(a) The circuit is drawn with a crossing. Is it planar? If so, redraw it without a crossing.

(b) Node voltage analysis: take node 4 as the datum. How many unknown node voltages are there? Write KCL, then $Y_n\,\mathbf{e} = \mathbf{i}_n$, and solve.

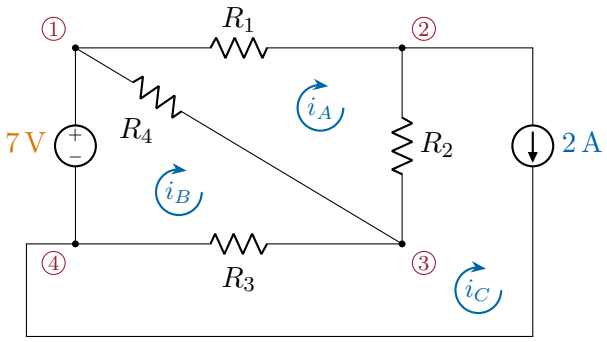
(c) Mesh current analysis on your new drawing. How many unknown mesh currents are there? Write KVL, then the mesh equations in matrix form, and solve.

(d) Find the current of R_4 and the power delivered by the 7 V source with both results. Do they agree?

Solution

Step a. Planar?

There are $n = 4$ nodes and $b = 6$ branches, and every pair of nodes is joined by one branch. The crossing is not needed: take the 2 A source around the outside. So the circuit is **planar**, and mesh analysis applies. It has $b - n + 1 = 3$ meshes.



Mesh A: R_1, R_2, R_4. Mesh B: R_4, R_3 and the 7 V source. Mesh C: R_2, R_3 and the 2 A source. All mesh currents are clockwise.

Step b. Node voltage analysis

Unknowns. The datum is node 4, so $e_4 = 0$. The 7 V source is between node 1 and the datum: $e_1 = 7\,\text{V}$ is **known**. Unknowns: e_2, e_3.

KCL (currents leaving the node):

$$\begin{aligned} \text{node 2:} \quad & \frac{e_2 - 7}{1} + \frac{e_2 - e_3}{1} + 2 = 0 \\ \text{node 3:} \quad & \frac{e_3 - e_2}{1} + \frac{e_3}{1} + \frac{e_3 - 7}{2} = 0 \end{aligned}$$

Matrix form. Move the known terms to the right:

$$\underbrace{\begin{bmatrix} 2 & -1 \\ -1 & 2.5 \end{bmatrix}}_{Y_n} \begin{bmatrix} e_2 \\ e_3 \end{bmatrix} = \underbrace{\begin{bmatrix} 7-2 \\ 3.5 \end{bmatrix}}_{\mathbf{i}_n} \quad (\text{S},\ \text{V},\ \text{A}).$$

By inspection: a diagonal entry is the total conductance at the node ($1+1$; $1+1+\frac{1}{2}$), an off-diagonal entry is minus the conductance between the two nodes. The right side holds the source currents into the node, and $G\,e_1$ for each conductance to the known node 1.

Solve. Row 1 gives $e_3 = 2\,e_2 - 5$; row 2 then gives $4\,e_2 - 12.5 = 3.5$:

$$e_2 = 4\,\text{V}, \qquad e_3 = 3\,\text{V}.$$

Step c. Mesh current analysis

Unknowns. The 2 A source is only in mesh C, in its direction: $i_C = 2\,\text{A}$ is **known**. Unknowns: i_A, i_B.

KVL (clockwise, voltage drops):

$$\begin{aligned} \text{mesh } A: \quad & 1 \cdot i_A + 1\,(i_A - i_C) + 2\,(i_A - i_B) = 0 \\ \text{mesh } B: \quad & 2\,(i_B - i_A) + 1\,(i_B - i_C) - 7 = 0 \end{aligned}$$

Mesh B goes up through the source, from $-$ to $+$: a rise of 7 V. Mesh C is not needed: its KVL only adds the unknown voltage of the current source.

Matrix form. Plug in $i_C = 2$ and move the known terms to the right:

$$\underbrace{\begin{bmatrix} 4 & -2 \\ -2 & 3 \end{bmatrix}}_{Z_M} \begin{bmatrix} i_A \\ i_B \end{bmatrix} = \begin{bmatrix} 2 \\ 7+2 \end{bmatrix} \quad (\Omega,\ \text{A},\ \text{V}).$$

By inspection: a diagonal entry is the total resistance of the mesh ($1+1+2$; $2+1$), an off-diagonal entry is minus the shared resistance (R_4).

Solve. Row 1 gives $i_A = 0.5 + 0.5\,i_B$; row 2 then gives $2\,i_B - 1 = 9$:

$$i_A = 3\,\text{A}, \qquad i_B = 5\,\text{A}.$$

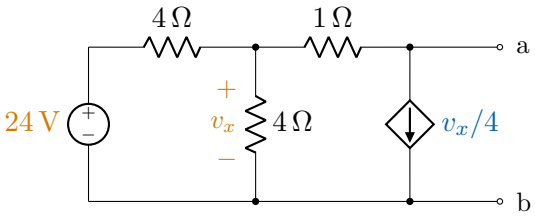
Step d. Compare

	Node analysis	**Mesh analysis**
Unknowns	2 (e_1 fixed by the 7 V source)	2 (i_C fixed by the 2 A source)
R_4, from 1 to 3	$(e_1 - e_3)/2 = (7-3)/2 = 2\,\text{A}$	$i_B - i_A = 5 - 3 = 2\,\text{A}$
7 V source, up	$\frac{7-e_2}{1} + \frac{7-e_3}{2} = 3 + 2 = 5\,\text{A}$	$i_B = 5\,\text{A}$

Both give the same answers. The source current is 5 A, leaving its + terminal: the source **delivers** $7 \cdot 5 = 35$ W.

Quick check (optional): R_1, R_2, R_3 carry 3, 1, 3 A and R_4 carries 2 A, so the resistors absorb $9 + 1 + 9 + 2 \cdot 4 = 27$ W. The current source has $e_2 - e_4 = 4$ V across it and absorbs $4 \cdot 2 = 8$ W: $27 + 8 = 35$. ✓

Question 2 — Thévenin, Norton and maximum power



The dependent source is a voltage-controlled current source $v_x/4$ (in A, with v_x in V).

(a) Find the Thévenin equivalent seen from the port a–b.
(b) Find the Norton equivalent.
(c) Check R_{th} with a test source. Why can it not be found by combining resistors?
(d) Which load resistor R_L at a–b receives the maximum power? How much?

Solution

Step 1. Open circuit: v_{oc}

With a–b open, the 1 Ω resistor carries exactly the dependent source current $v_x/4$ (KCL at a). KCL at the top node of v_x:

$$\frac{24 - v_x}{4} = \frac{v_x}{4} + \frac{v_x}{4} \quad \Rightarrow \quad v_x = 8\,\text{V}.$$

Then $v_{\text{oc}} = v_x - 1 \cdot \frac{v_x}{4} = 8 - 2 = 6$ V, so $v_{\text{th}} = 6$ V.

Step 2. Short circuit: i_{sc}

With a–b shorted, $v_a = 0$ and the 1 Ω resistor carries $v_x/1$. KCL at the top node:

$$\frac{24 - v_x}{4} = \frac{v_x}{4} + v_x \quad \Rightarrow \quad v_x = 4\,\text{V}.$$

KCL at a: $i_{\text{sc}} = v_x - \frac{v_x}{4} = 3$ A. So $R_{\text{th}} = v_{\text{oc}}/i_{\text{sc}} = 2\,\Omega$. Thévenin: 6 V in series with 2 Ω. Norton: 3 A in parallel with 2 Ω.

Step 3. Test source

Kill the 24 V source (a short) but keep the dependent source. Apply v at a–b. KCL at the top node: $\frac{v_x}{4} + \frac{v_x}{4} + (v_x - v) = 0$, so $v_x = \frac{2}{3}v$. The test current is

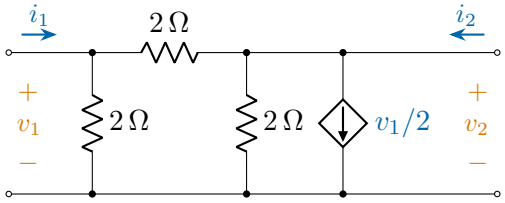
$$i = (v - v_x) + \frac{v_x}{4} = \frac{v}{3} + \frac{v}{6} = \frac{v}{2} \quad \Rightarrow \quad R_{\text{th}} = \frac{v}{i} = 2\,\Omega. \checkmark$$

Series–parallel combination ($4 \parallel 4 + 1 = 3\,\Omega$) gives the wrong answer: the dependent source is not killed and changes the current.

Step 4. Maximum power

$R_L = R_{\text{th}} = 2\,\Omega$, and $P_{L,\max} = \dfrac{v_{\text{th}}^2}{4R_{\text{th}}} = \dfrac{36}{8} = 4.5$ W.

Question 3 — a two-port with a dependent source



The dependent source is a voltage-controlled current source $v_1/2$ (in A, with v_1 in V).

(a) Find the conductance matrix $\mathbf{G}$ by short-circuit tests.
(b) Find the resistance matrix $\mathbf{R}$. Check r_{12} and r_{22} with an open-circuit test.
(c) Find the hybrid matrix $\mathbf{H}$.
(d) Does the transmission matrix $\mathbf{T}$ exist? Explain.
(e) Is the two-port passive? (Write $P_N = v_1\, i_1 + v_2\, i_2$ in terms of v_1 and v_2.)

Solution

Step a. G by short-circuit tests

Port 2 shorted ($v_2 = 0$): the left and the middle 2 Ω are both across v_1, so $i_1 = v_1/2 + v_1/2 = v_1$. At the port-2 node, the middle resistor brings $v_1/2$ in and the dependent source takes $v_1/2$ out, so $i_2 = 0$: $g_{11} = 1\,\text{S}$, $g_{21} = 0$.

Port 1 shorted ($v_1 = 0$): the dependent source is 0. Then $i_2 = v_2/2 + v_2/2 = v_2$ and $i_1 = -v_2/2$: $g_{22} = 1\,\text{S}$, $g_{12} = -\frac{1}{2}\,\text{S}$.

$$\mathbf{G} = \begin{bmatrix} 1 & -\frac{1}{2} \\ 0 & 1 \end{bmatrix} \text{S}.$$

Step b. R

$\det \mathbf{G} = 1$, so

$$\mathbf{R} = \mathbf{G}^{-1} = \begin{bmatrix} 1 & \frac{1}{2} \\ 0 & 1 \end{bmatrix} \Omega.$$

Check, port 1 open ($i_1 = 0$): KCL at port 1 gives $\frac{v_1}{2} + \frac{v_1 - v_2}{2} = 0$, so $v_1 = v_2/2$. KCL at port 2: $i_2 = \frac{v_2}{2} + \frac{v_2 - v_1}{2} + \frac{v_1}{2} = v_2$. So $r_{22} = v_2/i_2 = 1\,\Omega$ and $r_{12} = v_1/i_2 = \frac{1}{2}\,\Omega$. ✓

Step c. H

The first row of $\mathbf{G}$ gives $v_1 = i_1 + \frac{1}{2}\, v_2$; the second row gives $i_2 = v_2$:

$$\begin{bmatrix} v_1 \\ i_2 \end{bmatrix} = \underbrace{\begin{bmatrix} 1\,\Omega & \frac{1}{2} \\ 0 & 1\,\text{S} \end{bmatrix}}_{\mathbf{H}} \begin{bmatrix} i_1 \\ v_2 \end{bmatrix}.$$

Step d. T

$\mathbf{T}$ gives port 1 from port 2: $[\, v_1 \;\; i_1 \,]^T = \mathbf{T}\, [\, v_2 \;\; -i_2 \,]^T$. Its inputs are v_2 and $-i_2$.

Test: set both inputs to zero. Make port 2 dead: $v_2 = 0$ and $i_2 = 0$. Row 2 of $\mathbf{G}$ is $i_2 = v_2$, so this is allowed. Row 1 then gives $i_1 = v_1$, and v_1 can still be any value. A nonzero port 1 with a dead port 2: **$\mathbf{T}$ does not exist**.

Meaning. $g_{21} = 0$: port 1 has no effect on port 2. Port 2 behaves like a 1 Ω resistor, whatever happens at port 1. So watching port 2, we cannot tell what happens at port 1, and port 1 cannot be written in terms of port 2.

Rule. When $\mathbf{G}$ exists, $\mathbf{T} = -\dfrac{1}{g_{21}} \begin{bmatrix} g_{22} & 1 \\ \det \mathbf{G} & g_{11} \end{bmatrix}$: $\mathbf{T}$ exists exactly when $g_{21} \neq 0$. (In the same way, from $\mathbf{R}$: when $r_{21} \neq 0$.)

Step e. Passive?

$$P_N = v_1\,(v_1 - \tfrac{1}{2}\,v_2) + v_2\,v_2 = {v_1}^2 - \tfrac{1}{2}\,v_1\,v_2 + {v_2}^2 = \left(v_1 - \tfrac{1}{4}\,v_2\right)^2 + \tfrac{15}{16}\,{v_2}^2 \geq 0.$$

$P_N \geq 0$ for all port variables: the two-port is **passive**, even though it contains a dependent source.